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The Messy Reality of Agglomeration Economies in Urban Informality: Evidence from Nairobi's Handicraft Industry

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Summary. — This paper investigates intermediate input effects and labor pooling in the context of urban informality, examining the handicraft industry of Nairobi, Kenya. The paper tests if conditions found in informality interfere with transition pathways in production clusters. 102 semi-structured interviews were conducted with industry participants, including 69 production firms. Findings confirm the existence of important agglomeration economies, but those benefits are curtailed due to constraints of informality on firms, especially location constraints in urban space. At the same time, informality also introduces particular diseconomies of agglomeration that undermine the value of proximity of firms to each other, particularly within clusters.
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1. INTRODUCTION

Urban informality as defined herein represents a significant share of employment and housing in many developing world cities and is a frequent research topic touching on important aspects of economic development, urban livelihoods, and human development. Informality and, in particular, the informal economy, is often in the eye of the beholder (Potts, 2008) and has been described as a backward and nagging problem to be overcome by modern industry (Boeke, 1961); a collective act of heroic entrepreneurism and resistance in the face of an ineffectual state (De Soto, 2000); and as a last bastion for survival of the neglected poor of the world (ILO, 1972).

Another frequent research topic also related to economic development, urban livelihoods, and human development, is the effect of agglomeration economies in developing world cities. Scholars have increasing interest in fostering agglomeration effects in the developing world; often seeking to extend the benefits observed in "The Third Italy" (Bianchi, Miller, & Bertini, 1997). The particular agglomeration economies in the Italian context interested scholars because growth came from small- and medium-sized firms such as those which typically dominate urban regions in the developing world. This scholarship focused on industrial districts in the developing world and examined the assumed benefits of agglomeration (Murray, 1990; Nadvi, 1992; Schmitz, 1989; Storper, 1990). These studies led to a special edition devoted to the topic in *World Development* in 1999. This research also generated a literature exclusive to African industrial clusters (McCormick, 1999; McCormick & Oyelaran-Oyeyinka, 2007; Meagher, 2007; Murphy, 2006, 2007; Zeng, 2008). Further, it is argued in *The World Development Report* for 2009, with special emphasis on Africa, that policy makers and development practitioners have largely failed to foster agglomeration economies in the developing world and that policy makers should act with haste to correct these failures (World Bank, 2008).

This paper investigates the relationship between urban informality and agglomeration economies in Nairobi, Kenya. Urban informality is thought to produce a "messy reality" in local applications (Kudva, 2009, p. 1625) that can obscure a clear understanding of relationships, processes, and resource allocation. This study recognizes this messy reality and seeks to create a clearer explanation of the relationship between informality and agglomeration economies by looking at the role of intermediate input effects and labor pooling in the Nairobi handicraft industry, particularly in clustered and non-clustered firms. Research questions include: *What effects of these two agglomeration economies are observable in the industry? Are there confounding influences of informality on these agglomeration economies? If so, how do they work and how do they create problems for cluster development?*

The results of the study suggest that the reality is indeed messy with three dynamics operating simultaneously: (1) while the industry seems to experience important agglomeration benefits such as increasing returns to scale due to access to suppliers, customers, and workers within clusters; (2) those benefits seem to be curtailed due to constraints of informality on firms, especially location constraints in urban space; and (3) at the same time, informality also introduces particular diseconomies of agglomeration that undermine the value of proximity of firms to each other, particularly within clusters. This article concludes by suggesting that the dynamics found in this industry, if widespread in other informal industries may, in part, help explain why informal industrial clusters so often fail to evolve into higher order industrial districts with robust and endogenous agglomeration economies (McCormick, 1999; Van Dijk & Sverrisson, 2003).

2. THEORY

(a) *Urban informality*

The term urban informality, or informality as defined and used throughout this article, includes the suggestion by Lloyd-Evans that the informal sector is "a heterogeneous group of activities and employment relationships that share one common characteristic—the lack of legal recognition, regulation or protection" (2008, p. 1885), and also includes the

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recognition that informality goes beyond the preoccupation with economic production. Therefore, this definition recognizes that informality is an organizing logic that influences governing norms of urbanization (Roy, 2005), housing for much of the urban population in the developing world (Hansen & Vaa, 2004) and the social production of urban space (Kudva, 2009). Informality is understood more as a process of creating spaces of exclusion (Smiley & Koti, 2010) where urban citizens often live their daily lives on the border of legality and legitimacy. However, this concept of informality is so broad that it is difficult to make it operational. All of these characteristics of urban informality will manifest in very local ways, but this article focuses on informal economic production nested in the larger concept. Therefore, operationally, informality is here a function of government registration of firms in which the informal status of urban land and government institutions are considered when appropriate to explain interaction between economic production and informal urban processes and space.

(b) *Informality and agglomeration*

This article deals with two agglomeration economies: *Intermediate input effects*, the benefits accruing to upstream and downstream suppliers due to increased interaction and proximity; and *Labor pooling*, labor search costs savings due to a local concentration of skilled labor.

These effects are described by Marshall (1890) and Krugman (1991). Access to raw materials and intermediate input providers used in the production of final products will likely be important to any industry with several value-added layers and continuously changing designs and customer tastes. Also, access to customers for intermediate and final products are likely to generate "shopping externalities" and other efficiencies where proximity to other producers creates increasing returns to scale. Labor pooling will occur as the density of economic activity for an industry increases in an urban region and employees are drawn to areas where employment opportunities are readily available and employers benefit from a ready "pool" of labor for any present or future need. This effect is particularly important when specific skills are required by an industry. However, we must differentiate between localization economies and urbanization economies regarding both of these effects (Rosenthal & Strange, 2004). Urbanization economies accrue to large urban contexts where diversity of workers, business inputs, and employers creates a positive externality. Localization economies impact high concentrations of same or similar firms because similar firms require similar inputs and because employees with desired skill sets tend to pool near the employers who use them thereby benefiting both employees and employers. This article is primarily concerned with localization effects because it examines a single industry, or more accurately a series of sub-industries. These positive externalities tend to establish clusters of relatively dense agglomerations of firms in the same or similar industries.

The relationship between informal space and agglomeration economies is a research topic gaining increasing interest. Amin (1994) initially provides a skeptical perspective suggesting that informal firms will likely be *unable* to specialize in product development and labor force skills, generate innovations, or rely on "institutional thickness" in terms of available options for conflict resolution and collective action. Murphy (2006, 2007) suggests that informal Tanzanian furniture makers are impacted by proximity to main markets where certain benefits of agglomeration are present, but that spatial patterns expected by agglomeration theory are not fully observed primarily due to firm location decisions that prioritize safety and stability. Similarly, Sonobe, Akoten, and Otsuka (2011) find that prosperous metal workers in Nairobi must leave behind industry clusters in order to grow their business due to congestion and high rates of competition.

McCormick (1999) provides data from six previous studies of African industrial clusters in which she finds limited agglomeration effects due to the forms of competition she observes and the low entry barriers within the industry. The cases studied by McCormick and Oyelaran-Oyeyinka (2007) and Zeng (2008) largely establish that the institutional context undermines the ability of African clusters to source human capital, technology, and other important inputs required for robust agglomeration effects. Meagher's (2007) analysis of informal clusters in Nigeria shows that the combination of structural adjustment policies and state neglect tends to increase the economic pressures on firms causing existing informal networks to fragment into highly counterproductive practices such as design stealing, undercutting, and other types of pernicious competition.

Other studies contain more positive results that suggest informal industries can and do benefit from agglomeration efficiencies. Chirisa (2009) conducted a spatial study of informal production in Harare, Zimbabwe and found that the traditional orienting factors of agglomeration, such as the CBD and industrial zones, were highly relevant and determined spatial relationships for informal firms. Kimura (2011) and Chung (2012) found that ethnic or kinship ties are important conduits for transfer of skills and upgrades to technology in informal clusters. Kinyanjui and Khayesi (2005), Lyons and Snoxell (2005) and Kinyanjui (2010) found that a considerable amount of social capital is created in informal clusters in which firms rely on each other in difficult times to overcome formal institutional weakness. Ali, Peerling, and Zhang (2010) suggested that the lowered entry barriers in informal clusters are a positive externality that allows firms to overcome financing shortfalls. This agrees with the ILO's (1972) long-standing position that informal firms and clusters offer important life-sustaining livelihoods and thus deserve material and policy support.

There is an important dichotomy between the studies that are more positive about agglomeration effects in informal production and those that are less positive. Among *formal* firms, low-tech industries with heavy labor inputs are often cited as a key to successful national development strategies. These industries greatly benefit from healthy agglomeration economies (Fan & Scott, 2003; Scott, 2006). This is especially true of the two effects considered here. In these studies, robust labor pooling and intermediate input effects are important to understanding the success of Asian development strategies. The low-tech/labor-intense qualities of these incipient industries are also definitional aspects of informal industries such as the ones examined in the above cited literature and in this article. As much of the cited literature suggests, there is reason to doubt that agglomeration economies can achieve the same results in informal industries as in formal industries.

However, there is reason for hope that informal production clusters can benefit from agglomeration efficiencies over time. An important paper by Van Dijk and Sverrisson (2003) provides a typology of production clusters in the developing world and mechanisms for transition within clusters. The authors define clusters as "relatively dense networks of enterprises and organizations, the value chains of which are connected but not necessarily through what we usually understand by economic transactions" (p. 185). The authors

establish five types of clusters: *Location Clusters* are basic level clusters, usually of diverse informal firms, in which information exchange is the main benefit and imitation is the prevailing technological method. *Local Market* clusters are areas in which similar firms gather for easy access to customers and inputs. In these clusters competition is intense and leads to product development to enable firms to stand out from their competitors. *Local Network* clusters are clusters in which firms, through the evolutionary process of product development, create important levels of specialization by the resulting division of labor. Production technology also evolves into complimentary production processes. *Innovative Clusters* are clusters that have developed the ability to adapt and reverse engineer products in novel ways. *Industrial Districts* are clusters in which co-operation and collective innovation exist along with productive forms of competition.

This typology considers clusters as evolutionary processes whereby it is possible for lower order clusters to move toward higher order operations. The transition from the lowest order (Location Clusters) to the second order (Local Market) requires facilitation of basic agglomeration efficiencies as co-locating firms begin to draw materials, workers, and consumers to its location. This in turn brings more firms that wish to co-locate to take advantage of the apparent increasing returns. The transition from the second order (Local Market) to the third (Local Network) requires increasing labor specialization and complimentary production processes in addition to the other agglomeration effects present in the cluster. Movement toward the forth (Innovative Clusters) and fifth (Industrial Districts) involve even greater depth of specialization and flexibility by firms as well as a growing ability to foster collective action.

The requirements of transition within this typology may in part explain why the findings in studies of agglomeration economies in informal clusters are inconsistent throughout the literature. Studies looking for basic agglomeration effects, such as intermediate input and labor pooling, are likely to find evidence in existing clusters. Depending on the clusters' evolutionary status, those looking for higher order effects may not find them.

The hypothesis of this article is that conditions often inherent in informal spaces and production interfere with the mechanisms of transition that allow incipient clusters to evolve. In particular, the mechanics of lower order agglomeration effects (intermediate inputs and labor pooling) undermine the stability and trajectory of production clusters when working in conditions of informality. This is because lower order agglomeration effects may function in certain ways within informality that prevent evolution to higher order agglomeration economies. Research questions include: *What effects of these two agglomeration economies are observable in the industry? Are there confounding influences of informality on these agglomeration economies? If so, how do they work and how do they create problems for cluster development?* The pernicious forms of competition and business practices, constrained spatial patterns, and the problems related to entry barriers and the oversupply of labor in the cited literature will be investigated for their impact on the potential for agglomeration evolution.

3. INDUSTRY INFORMATION AND STUDY AREA

Nairobi is an important center for handicraft production in East Africa. Though production of handicrafts is located all over the city, it also is highly concentrated in select production clusters. Several sub-industries are represented including: bone art (primarily bead making), banana leaf art, batiks, wood carving, soapstone carving, sandal making, textiles, and beaded jewelry design and assembly. The sub-industries are together treated as one larger handicraft industry and share retail markets, customers, a labor pool, and the same outlook on hiring and competition. There are several clusters throughout the city and they are generally classifiable as either production or retail clusters. Production clusters are often sub industry and ethnic group specific. Examples are the wood, bone, and soapstone production clusters. These clusters are dominated by the Kamba, Luo, and Kissii ethnicities respectively. Clusters can also be a mix of industry and ethnicity. For example, the Kariokor Market is an epicenter in the city for clustered production of baskets, woven bags, jewelry, and sandals and is a mix of many ethnic groups. Retail clusters are both stationary and itinerant. Stationary clusters include the Westlands Curio Triangle, the Embakasi shops, and the CBD, which includes the City Market and several higher end shops. Maasai Markets are itinerant markets that take place in different locations and on different days throughout the city.

It is estimated that Kenya's handicraft sector grew by approximately 70% during 2001–07 and was worth over US$ 12 Million in 2007 (Waweru, 2010). Like most informal industries, official statistics are not kept on the handicraft sector in Kenya which makes accurate estimates of the size and structure difficult. The International Labour Organization (2002) estimates that approximately 72% of all nonfarm employment in Kenya is informal. Since 1993 over 18,000 individual informal handicraft firms have joined the National Jua Kali Employment and Marketing Association (NJEMA), a marketing organization for handicraft firms. Approximately 600 firms, mostly from Nairobi, are currently active in the organization's activities (E. Emojong, personal communication, August 8, 2011). There is no way to know the exact size of the industry, but it is reasonable to conclude that it accounts for a sizeable source of employment and livelihoods in Nairobi.

4. METHODS

Between July and November 2011 in-depth semi-structured interviews were conducted with 102 respondents who were all participants in the Nairobi handicraft industry. The interviews were conducted in and around Nairobi, Kenya. The study focused on manufacturing firm managers/owner operators ($N = 69$), but included industry laborers ($N = 10$), retail traders ($N = 16$), and raw materials suppliers ($N = 5$). The majority of respondent firms (75%) were informal in that they were not a formally registered entity with the Kenyan Attorney General's Office; however, most firms had some type of tenuous governmental legitimacy established through various City Counsel arrangements which allow informal firms to occupy urban land.

Firms were identified for interview in a two-part snowball-styled process. In the first part, nongovernmental organizations (NGOs) that work within the handicraft industry were contacted and three such organizations provided a list of names and mobile phone numbers for the firms with which they work. Firms that agreed to be interviewed were also asked to provide an introduction to their suppliers, buyers, and competitors. In the second part, the visible retail and production centers throughout the city provided further firms. These locations represented important entry points which

allowed the researcher to approach proprietors for the study. After interviewing willing respondents chosen at random in these locations, these respondents were also asked to provide introductions to their suppliers, buyers, service providers, and competitors in a further snowball-styled sampling process. Additionally, the researcher performed a significant amount of direct observation of firms conducting business and was therefore able to spend substantial time in handicraft clusters throughout the city.

Four distinct organizations/spatial patterns for firms involved in the handicraft industry were observed in the study. Table 1 shows the proportion of these four patterns in the research sample for producers only ($N = 69$).

(a) *Urban home-based enterprise (HBE)*

Urban HBEs are businesses based in the home of the firm manager or owner operator. Urban HBEs are commonly based in Nairobi's informal settlements, but a few are operated out of formal residential neighborhoods. These businesses are spatially isolated from other handicraft firms in the sense that respondents did not identify other handicraft businesses in the immediate vicinity of their home.

(b) *Peri-Urban home-based enterprise*

Peri-Urban HBEs are firms based in the home of the firm owner and located in the urbanizing outskirts of the Nairobi metro region. The main difference between Urban and Peri-Urban HBEs is the significant commuting distance from the city center. This becomes important in terms of the separation between Peri-Urban firms and concentrations of buyers, labor, and other handicraft firms.

(c) *Clusters*

Adopting the cluster definition of Van Dijk and Sverrisson (2003) cited above, clusters are groupings of handicraft firms in an industrial or commercial area outside the home of the owner operator. All handicraft clusters in Nairobi have several similar characteristics. They are major centers of employment and many easily provide livelihoods to hundreds of individuals. Activities in the clusters are performed in very close proximity and mostly in the open. As will be discussed below, most clusters exist outside the private urban real estate market so that firms in clusters occupy the land in various ways other than freehold title or formal lease.

(d) *Isolated industrial*

Isolated Industrial firms are those that are located outside of the home of the owner operator in a commercial or industrial area, but not in any identifiable handicraft cluster. Though these businesses may be in a concentrated area of other light industries, they are not self-reported to be located among other handicraft firms.

The following map, Figure 1, depicts important locations in Nairobi for the handicraft industry. Urban HBEs, Peri-Urban HBEs, Clusters (production and retail), and Isolated industrial firms are indicated.

Table 1. *Spatial Pattern Proportion in the Sample*

Firms by location type for producers ($N = 69$)	
Urban HBE	24.64%
Peri-Urban HBE	20.29%
Cluster	42.03%
Isolated industrial	13.04%

The interviews produced firm level data on firm characteristics, production chain relationships, cost structures, etc., in a manner similar to the World Bank African Enterprise Surveys conducted in select locations throughout Sub-Saharan Africa since the 1990s (Bigsten & Söderbom, 2006). However, the questions were asked in a qualitative manner in such a way that each interview was unique and allowed the researcher to follow topics of interest in conversation style. As a result, while all respondents were questioned about firm characteristics, production, hiring decisions, etc.; the researcher could engage in lengthy follow-up exploration on topics including but not limited to cost structures, innovation, investment decisions, business climate, and relations with institutional authorities such as the police, courts of law, and representatives of the government.

All interviews were recorded and coded using qualitative data processing software for important contextual statements and cause and effect logic employed by respondents from their own perspective. The data collection method and the sample size do not allow for truly robust quantitative analysis or highly generalizable conclusions to all manifestations of informality. These are common constraints on informality research. However, the intent here is to build useful concepts and plausible theory explaining how informality and agglomeration economies interact. Therefore, basic quantitative data are used to suggest spatial relationships and summary statistics are used intermittently to summarize the experience of respondents. These are largely intended to provide context and corroboration for concepts generated by respondent quotes, which are used as the main explanatory evidence.

Important characteristics of respondents in the sample are included in Table 2.

5. FINDINGS

(a) *Intermediate input effects*

Through direct observation and respondent interviews it is apparent that most producers depend on small-batch middle men for access to raw materials (wood, hides, sheet metal, bone, etc.) or for intermediate manufactured goods (glass beads, paint, glue, and fabric). This is because bulk buying is cost prohibitive for most, or interaction with formal suppliers of industrial inputs is not possible due to the informal status of firms. All clusters were observed to contain small-batch middle men operators that provide inputs to producers. Clusters are often the only places to find needed materials in the quantities demanded which leads even nonclustered producers to enter clusters to find inputs.

Also, a product order of any significant size is often beyond what an individual owner operator can accommodate. As a result, firms frequently use subcontractors to fill orders. In the clusters, there is often an abundance of subcontracting opportunities within shouting distance which is a draw to customers and a benefit to clustered firms because the capacity constraints on individual producers are well known to traders and retailers. Placing an order with a cluster firm means the subcontractor to the firm receiving the order may well be standing next to him or her when the order is received. Therefore, search time for subcontractors is negligible and trader and retailing respondents confirm that larger orders can be filled more quickly by clustered firms than with isolated firms. Additionally, it is common for buyers to order more than one

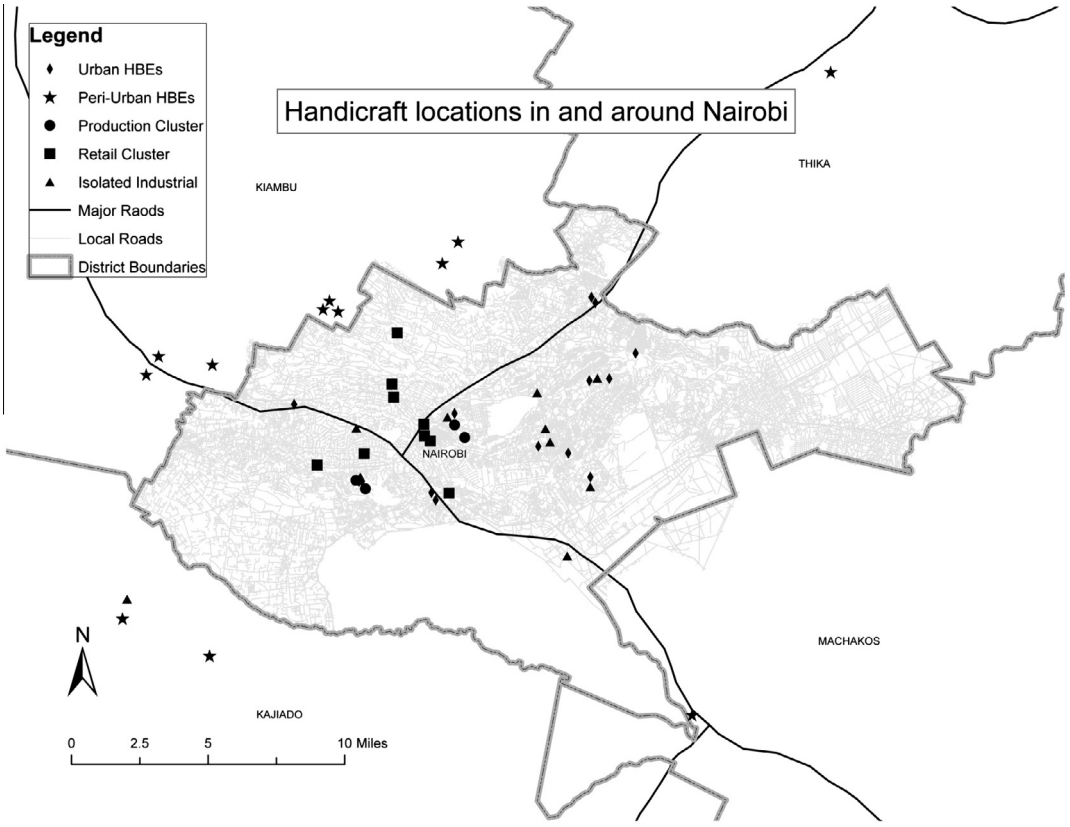


Figure 1. *Important handicraft locations in Nairobi.*

Table 2. *Characteristics of Respondents*

Characteristics	Registered	Unregistered	Range
Male	63%	63%	
Female	37%	37%	
Age	45	38	20–70 years
Average education	Complete secondary	Some secondary	None to university
Years in industry	16	15	0–55 years

kind of item from a firm. The originating firm may or may not have experience producing all the items. Within a cluster, finding products to fill the order is easier because there are often hundreds of experienced producers of all types of items.

Though the handicraft industry is export oriented, marketing is generally a significant challenge. Outside of the clusters, there are few good ways for individual producers to connect to overseas markets or to tourists, and few good ways for those purchasers of goods to connect with the producers. Buyers and sellers have to interact face to face to conduct business. This is especially true during initial dealings so that often they must be physically present or one will not know the existence of the other. Therefore, clusters not only reduce the search costs for buyers and sellers, but in many cases they are essential for the industry's existence.

These are powerful agglomeration effects and are summed up by Respondent MD's comment referring to the jewelry industry at the Kariokor cluster. She states the opinion of many firms of the benefits of working in a cluster when she says: "There is only one place to do this work."

Each respondent was asked to identify important factors contributing to their location decision. Respondents were given a list of possible responses and could identify more than one as important. Table 3 is a summary of their responses displayed as the percentage of firms identifying individual factors as important.

Proximity to suppliers and customers, two factors associated with intermediate input effects, are included in the top three location factors for firms in the sample. The importance of these effects becomes clearer when the decision factors are broken down by firm location categories for producers only (*vs.* retailers, laborers, etc.). Table 4 shows the percentage of producer firms identifying proximity to customers and proximity to suppliers by owner operator location type. Taken together the benefits of proximity to suppliers and customers are especially important to clustered firms in their location decisions and these differences are statistically significant. The relatively low number of HBEs and Isolated Industrial firms identifying these factors as important does not necessarily suggest that access to customers or suppliers is unimportant, but suggests

Table 3. *Percentage of Firms Identifying Important Location Factors*

Proximity to customers	51%
Cost of location	39%
Proximity to suppliers	25%
Proximity to employees	24%
Other nonbusiness reasons	19%
Social reasons	14%
Increased security	11%

Table 4. *Intermediate Input Effects as Location Factors, Producers (N = 69)*

Location type	Customers important*	Suppliers important*
Urban HBE	6% ($N = 1$)	0% ($N = 0$)
Peri-Urban HBE	7% ($N = 1$)	29% ($N = 4$)
Cluster	66% ($N = 19$)	52% ($N = 15$)
Isolated industrial	44% ($N = 4$)	22% ($N = 2$)

*A chi-square test of homogeneity confirms the dissimilarity of these responses across the categories at the .05 level.

that other factors such as cost were more important in location decisions.

(b) *Intermediate input benefit constraints*

The research recorded the origin histories of the handicraft clusters in Nairobi. Histories were constructed from recollection of respondents as well as available corroborating sources. In summary, every handicraft production cluster in Nairobi requires some form of government intervention to exist. In many of the clusters, the government must directly allocate land to be occupied by the cluster activities. The wood carvers co-op at Gikomba, the Kariokor Market, the bone cluster at Kibera Hawker's Market and the Maasai Market all have recognized permission to occupy publicly controlled land even though very few of the inhabitants are recognized commercial entities registered with the Kenya Attorney General's office. In other cluster examples, the government must tacitly approve occupancy of the land associated with the cluster or the cluster will be disbanded through eviction. Eviction is an almost daily occurrence for informal or unregistered land uses in Nairobi. Thus, many cluster locations are, essentially, government-sanctioned squatting for which the approval can be removed at any time. This is true for the Westlands cluster, the bone cooperative on the Kibera railway, and the Kigali Market, whose demise discussed below demonstrates the problem with relying on government-fostered proximity for informal industries. In each of these cases, cluster activities are allowed through government intervention in the form of recognizing a cooperative or tacit approval of the cluster by issuing physical occupancy licenses to erstwhile unregistered trespassers. Both of these forms of government intervention, whether tacit or overt, is difficult to obtain and keep.

Government intervention is required for informal clusters to exist because there is no easy way for firms to group together independently or organically. If all things are equal, firms could group through the urban real estate market by paying rent or purchasing land in the private market. In informal industries, most of the actors are priced out of the formal urban real estate market or, through their informal status, are ineligible to participate as a business or legitimate economic actor. However, clustering is difficult even in the informal real estate market, where land occupancy transactions take place on land with insecure title or tenure. Even if government entities approve or ignore clustering activities, land rent practices in the informal real estate market make clustering difficult to achieve and likely unsustainable over time.

Respondents engaging the informal real estate market explain that without any kind of formal lease or protections, the landlord can arbitrarily raise the rent from month-to-month if the landlord believes an individual business has cash on hand to pay higher rents. They will also raise rent when firms begin to gather together making proximity to the cluster valuable. Rental value is not based on general market conditions such as how much the landlord could charge another renter, but is based on how much can be extracted from the current occupant before the occupant leaves. For example, respondent BO described problems with informal landlords as follows: "Sometimes they over estimate your income, he thinks you are making a lot so he wants you to pay more than what you are paying."

While this behavior is possible in formal real estate markets globally, in the informal real estate market, the absence of any institutional protections enables such conduct in real time. A good example is the experience of respondent BO quoted above. BO's work in the Kibera informal settlement attracted attention from a UN agency which visited one day accompanied by a significant UN security detail. This visit drew much attention from BO's neighbors. The next day, the landlord visited BO's workshop to demand immediate payment for BO to remain on the premises and raised all future monthly rent payments.

Proximity to other firms is difficult to achieve for informal firms outside of tacit or overt government intervention due to capital constraints or illegitimacy. While government intervention allows proximity which fosters the agglomeration effects observed above, governments are fickle and favor can be removed at any time without institutional protections. As an example, Kigali Market no longer exists, but was mentioned repeatedly by respondents. This was once a cluster in the heart of the city occupied by handicraft retailers and producers through government tacit approval. The municipal government decided to sell the land in July 1998, the market was bulldozed and the remains of hundreds of livelihoods burned in the middle of the night (Ngunjiri, 1998).

In another example, Kariokor Market which is described above as a cluster for several handicraft sub-industries, has been in existence since the 1920s. Despite its longevity, respondent ES refers to it as follows: "This is the City Council's market. They can take it back whenever they want." Even though the market has been in operation for over 90 years, the experience of informal firms with municipal matters makes them permanently treat their business arrangements as temporary. Respondents regularly report that the insecurity of their tenure changes the way they would otherwise do business even when they have been in the same location for years. They do not invest in their place of business and see no logic in building structures from which they may be evicted.

(c) *Intermediate input diseconomies*

(i) *Subcontracting*

As stated above, the industry generally relies heavily on subcontracting and subcontracting appears to be a powerful intermediate input externality in clusters. The practice is widespread for cluster firms. The average owner operator relies on 5.5 subcontractors and is a subcontractor to 4.7 other firms. Conceptually, this strategy is likely a low-cost alternative to cost prohibitive vertical integration because firms can rapidly expand production capability without taking on costs

associated with expanding internal capacity. It is important to note that subcontracting in the industry is not largely for intermediate inputs, *per se*, but for consignments of finished product from erstwhile competitors. While this is an important positive externality, it creates or fosters a number of problems including adverse selection and problems with proprietary designs, which in turn represent important diseconomies of agglomeration within clusters. These problems are now discussed.

(ii) *Adverse selection within clusters*

Adverse selection is a problem in asymmetrical transactions (Kreps, 2004). When the seller or subcontractor has the essential information regarding quality or competency and the buyer has little such information, the seller is free to charge a high quality price for a low-quality good. When no single producer has an incentive to produce higher quality items, the entire industry will tend to produce lower quality, low-cost items (Akerlof, 1970).

A common complaint received particularly from cluster-based respondents is that their work is constantly undercut in quality and price by their surrounding competitors or their own production is made problematic by adverse selection conditions with their own subcontractors. Within handicrafts this can be common because production requires several layers of value addition. At each step cheaper materials (dye, paint, glue, leather, bead, and varnish) or processes (for example kiln drying *vs.* fireside drying of wood or clay) can be used in the place or more expensive higher quality options. The corner cutting may not impact a handicraft consumer, trader, or originating firm until after the point of sale.

As an example, EK, a cluster owner operator, described the common difficult choices faced by firms. She wants to keep her costs down and expand her production capability which requires she hire subcontractors. Unfortunately, she knows that this often costs her more in the long run because she gives up control of her own production process resulting in common delays in delivery and frequent problems with low-quality production. She must often use subcontractors that she knows little about. She describes the precariousness of her position as follows: "You know if they know how to do [the work] or not when they bring back the product."

The typical way formal firms overcome conditions of adverse selection is though institutional protections such as contract enforcement and government regulation, through business strategies like long-term relationships, or though industrial organization realities such as effective barriers to new entrants (Kreps, 2004).

In the handicraft industry in Nairobi, as with many informal industries, there is little or no regulation of production and institutional protections are thin. All respondents were asked if they would seek redress from courts of law or the police if they were victims of theft or if they were cheated during a business transaction. In the sample, 47% of registered formal firms have confidence in their access to government institutions, but only 19% of unregistered informal firms have confidence that government authorities will help if a problem arises.

Other methods of overcoming adverse selection seem similarly unavailable to informal firms. Under formal conditions of adverse selection, customers eventually discover problems and take their business elsewhere and the corner cutting operator will go out of business (Kreps, 2004). In informality, this is likely complicated because nearly unlimited numbers of firms can replace an existing firm due to minimal start-up costs and the low market power of the typical firm. This will be discussed further in the context of labor pooling.

It is quite common for nonclustered firms to indicate that cluster production quality is a major problem. This suggests that adverse selection creates a known negative externality. Respondent GC was a handicraft dealer with a formal retail shop in Nairobi's CBD. He buys from both cluster and noncluster dealers. He explained that his own buying strategy is to only source from the clusters when low-quality products are sufficient to fill an order for tourist consumption. While this may suggest a niche that is being properly exploited, it also functions as a diseconomy likely impacting the trajectory of the cluster.

As an example, Respondent CO is an intermediary for international buyers who organizes local producers for large orders. He spends time mentoring individual businesses and the main advice he gives quality producers is to leave the clusters as soon as possible. He stresses that a reputation for quality, which is the only way to secure lucrative international orders, is difficult while located in a cluster. In addition, noncluster producers often cite quality distinctions as a major reason for moving to a location outside of the cluster.

There are two strategic responses to adverse selection not mentioned above that help explain this behavior. These are "signaling" and "screening" (Kreps, 2004). Signaling occurs when a firm volunteers some kind of information that tips the buyer to the kind of product offered; alternatively, screening occurs when a buyer demands certain information from a seller that will screen the seller as to producer type. Both seem to take place in the informal industry; however, rather than the signal or screen being a pricing mechanism or imparting other information, the signal is often the *location* of the producer. That is, firms must (re)locate outside of clusters to signal the quality of their products and international buyers may be screening producers to those located outside the clusters. This has significant implications for cluster development if the higher quality firms of clusters must relocate for profit maximization. Adverse selection creates pernicious effects for business transactions, which are in turn a potentially significant diseconomy of agglomeration. Common remedies employed by formal firms are largely unavailable, and the most effective remedy open to informal firms is to signal to customers a higher probability of finding quality items by avoiding close proximity to other informal firms.

(iii) *Proprietary designs and informal proximity*

Cluster producer Respondent MD, summarized a common strategy for her and others in the industry: "I will see what you are making and I will copy." Informal firms in the industry steal the designs of competitors without apology. This includes passive imitation, but also more active replication. As an example, clustered respondents describe that it is common for competitors to approach a buyer in the act of ordering a new design from a producer and offer to produce a prototype of the same design for mass produce at a lower cost.

Respondent SD and her partner made African dolls and other textile items. They secured a spot at an international exhibition of craft items in Nairobi. They developed new designs specifically for the exhibition that they were confident would not only sell, but lead to future orders from international buyers. To produce the amount of product they expected to sell, they enlisted a competitor as a subcontractor. The subcontractor filled the requirements of the order, but when SD arrived at the exhibition, the subcontractor had also secured a space and was selling the very same designs SD had developed and produced by the subcontractor, who had in turn, subcontracted out the same design.

Design copying is a practice common to most handicraft firms because there are few secrets and essentially no institutional protections and the conditions of production in the clusters gives greater opportunity to copy. The production premises of informal firms are normally open air. The local term for these types of business is "jua kali" which literally translates to English as "hot sun" because most activities are done under the sun. The Respondent JJ operated a beaded jewelry production business in an isolated industrial location. He was formerly in a jewelry cluster. He moved because he felt that locating in the cluster gave him no control over proprietary designs. "I hate copycats, I hate copycats," was the summation of his thoughts on the matter.

(iv) *Informal competition as hypercompetition*

Hypercompetition is a condition in which increasing competition and constant undercutting tactics make markets volatile and keep any firm from gaining and holding market shares. Hypercompetition suggests that it may be impossible for firms within the same industry to maintain competitive advantage for any significant time under certain circumstances (D'Aveni, 1994). Hypercompetition and other forms of pernicious competition are observed in African industrial literature cited above (McCormick, 1999; Meagher, 2007). While hypercompetition is not directly related to subcontracting *per se*, as will be discussed, it does seem to be related to proximity created by basic agglomeration effects and is therefore discussed here.

The nature of competition was a common issue for respondents in the study. Respondent SC is a formally registered sandal maker in a predominantly informal cluster. He described his experience as follows:

> "Here (the cluster) you can't grow because there are people here that are selling at a loss just for the sake of selling. . .they don't budget. . .he just wants to get what he can eat today."

Respondent PD had less than a year's experience when he described a common sales practice: "Most of us get to desperation. And if you are not selling you will sell it at a loss just to eat."

In this context, firms are continuously faced with relentless downward pressure on prices. Recent research in other parts of Africa, suggest that in several locations 60% of informal manufacturing firms are "opportunity" firms and 40% are "necessity" firms. Opportunity firms enter industries because of perceived opportunity and necessity firms engage for survival (Amin, 2009). This is likely a diseconomy particular to informality in which opportunity firms are in competition with others essentially seeking an urban subsistence manufacturing livelihood. Acs and Varga (2005) suggest that the proportion of opportunity firms within the overall number of entrepreneurs impact overall economic growth. As suggested by Respondent SC above, this diseconomy is likely magnified through increased density of firms in clusters.

(d) *Labor pooling*

The dominant organizing logic in the handicraft system is the near ubiquitous reliance on "casual labor." The casual labor system is the widespread use of day laborers instead of permanent full-time employees. Usually, an employer hires as many day laborers as is needed for current orders and most often pays these laborers a daily piece rate. There is no long-term commitment by the employer to the casual worker and the casual worker is free to take on more than one employer in order to increase the probability of regular wages. The length of the relationship is dictated by the employer's cash flow and consistency of work.

Only 16% of the owner operator respondents in this sample indicate that they hire workers on a full-time permanent basis. The overwhelming majority of firms in this sample rely on casual workers for their labor needs. In Table 3 we see that proximity to employees is the fourth most commonly identified important location factor in the sample. Registered firms report hiring on average 4.5 casual workers in a given week and unregistered firms report about three casual workers per week.

When categorized by location type as in Table 5, there is no clear picture. While clustered firms suggest access to labor is relatively important so too do firms in other locations. The dissimilarity between these responses is not statistically significant.

However, interviews suggest that proximity to employers is very important to casual workers because, though not impossible, it is more difficult to find work in dispersed firms located outside of the cluster and opportunities to receive training are bountiful in the cluster. This is supported by the original understanding of Marshall that labor pooling creates a positive externality for workers due to lowered risk of unemployment. The clusters also provide important avenues for casual labor career advancement. In this sample, 60% of all current owner operators and 65% of all unregistered owner operators began as industry casual workers. Though this number is a clear majority, it does not include those that learned the business by "coming along side" friends and family members which is a common practice, particularly among women.

Casual workers tend to quickly transition to subcontractors and owner operators. As PD summarized his experience as a casual worker in a cluster before he started his own business: "We weren't actually going to work; we were going to work to learn something. Once you learn you go now on your own."

In this sample, casual workers reported average monthly earnings equivalent to about US$ 60. In comparison, the average monthly earnings for an unregistered owner operator were about US$ 275. Therefore, while the location decision responses from firms do not create a clear picture regarding labor pooling in terms of much higher response rates from clustered firms, from the perspective of casual workers, the agglomeration effect is likely an important positive externality.

(e) *Informal labor relations as diseconomy*

In the handicraft industry, most all sub-industries require some level of craftsman experience or specialty knowledge. Gaining skills in some sub-industries, like woodworking, can require years to master. Others skills, like jewelry design and assembly may take far less time to gain. However, in all cases workers with experience in the industry require far fewer resources from firms in terms of time and investment for increased productivity. For the purposes of this research, a skilled worker is considered one with experience in the industry that would presumably be more productive than an unskilled or untrained worker.

Table 5. *Producers Identifying Proximity to Employees as an Important Location Factor*

Location type	Employees important
Urban HBE	18% ($N = 3$)
Peri-Urban HBE	29% ($N = 4$)
Cluster	41% ($N = 12$)
Isolated industrial	44% ($N = 4$)

*A chi-square test of homogeneity cannot rule out the similarity of these responses with significance.

Respondent JK summarized the common outlook of firms toward labor decisions. When faced with the choice to hire a skilled worker who can join a firm's production process without training, or hire an unskilled worker who has to be trained, the decision is easy. Even though she acknowledges that skilled or experienced workers are more productive and could enhance her business in the long run she always trains new recruits because: "Training is free."

For casual workers there is a probationary time period when a new worker is not paid or receives only nominal wages while being trained to do the work. JK, a jewelry and sandal maker, explained it is far less expensive to bring in new recruits to the industry and train them to fill the order. During the initial time they are working on the order or, perhaps, until the order is filled, the trainees are not paid regular wages for the work because they are "in training." JK said that though there are workers all around her that do not need training and could complete the order at a much faster pace, she will only hire a skilled worker "if it is urgent." Essentially, JK is compensating the workers with skill acquisition.

This practice was found in all sub-industries of handicrafts production in Nairobi including sub-industries where skill acquisition takes significant time. For instance, even in woodworking, casuals are taken on to do basic sanding and cutting for a probationary period without wages and workers will often agree to new probationary periods if it will result in new skill acquisition.

This arrangement may not be problematic in and of itself, assuming casuals believe their training and skill acquisition was an adequate compensation for their labor; however, as will be discussed, when combined with other realities, pernicious diseconomies develop. One important aspect is that there is a constant flow of workers using the industry as livelihood refuge. Respondent KP discusses his own attitude toward introducing people into the industry as follows: "If a relative does not have employment, you know, you say—come let me show you what I do."

Respondent GC also stated "If a child does not want to go to school or someone loses a job they think they can always make the handicrafts."

The ILO (2002) states that 72% of all employment in Nairobi is informal which suggests limited amounts of formal employment. Kenya has an urban growth rate of 4.2% while the national growth rate is 2.2% (CIA, 2012). As a result, there is an endless stream of migrants and young urban workers entering the informal workforce.

In this sample, 78% of unregistered producers do not hire skilled workers. Instead they choose to train new workers they employ because of the cost savings and near unlimited demand for skills and livelihood acquisition. Firms constantly provide training to new entrants rather than utilizing existing skilled labor thereby creating a surplus of skilled or at least experienced workers. Conceptually, a surplus of skilled labor competing for existing jobs should have a depressing effect on wages. This would make skilled labor more affordable to existing firms thereby constraining the practice of training new workers because the existing skilled workers, now cheaper, are presumably more productive. The labor market should tend toward equilibrium regarding wages and number of workers. However, informality works against this. First, despite the depressing force of surplus skilled labor on wages, a skilled laborer cannot negotiate wages when his competitor works for free. Second, skilled laborers do not long remain workers but as described above, soon after acquiring necessary industry skills begin their own businesses. Essentially, skilled labor equals new owner operators. A skilled labor surplus quickly transforms to a surplus of competition which leads to a dearth of available skilled labor and a vicious cycle.

While this may resemble volatility seen in flexible labor markets in formal industries such as Silicon Valley (Benner, 2002), the skills and /or capital required by a formal start up that spins off is an effective barrier to entry constraining the overall amount of potential competition. Also, in formal contexts, markets are likely to tend toward equilibrium because competitors are unlikely to undercut their competition at a loss repeatedly for the sake of daily bread. This suggests that this kind of pernicious effect may be unique to informal labor markets.

The effect on firms is highlighted by the way some firms adapt to labor market conditions. NK runs a handicraft firm out of a Peri-Urban HBE. He makes keepsakes out of dried banana leaves. He stated that his most important factor for success in his business is that his location has a labor supply cut off from the larger pool. He stated that if he were located closer to the heart of the city, for instance in a cluster, he could not be as successful. In his current location, his workers do not have easy access to buyers except those who do business at NK's home compound. Therefore, going it alone would be difficult. Furthermore, the employees do not have access to other banana producers who would hire them as casuals or as subcontractors. As a result, though the start-up costs for a banana art business are negligible, his workers are unable to make products faster or cheaper than NK and there are no competing producers to offer such opportunities. NK is protected to a degree from hypercompetition and can afford regular wages for his employees, effectively ending the vicious cycle on a local scale. His employees remain in long-term employment with their regular, if meager, wages. There is far less incentive or opportunity to become owner operators.

This is not intended as a commentary on the fairness of the arrangement with NK's workers, but it does show that informal labor effects create powerful diseconomies of agglomeration that repel a number of firms. Firms that strategically avoid the negative externalities of the local labor pool must sequester their labor from the larger informal labor pool. Examples from the sample include firms that deliberately constrain their labor supply through a location (Peri-Urban areas), personal relationships, such as only hiring from a church group, or through other specific business strategies such as offering a set of young mothers flexible hours in exchange for long-term loyalty. While these may be novel and even positive strategies, they underline the larger diseconomy in the informal labor market.

6. DISCUSSION

Several points from the findings deserve additional attention. First, all notions of agglomeration economies are proximity dependent (Krugman, 1991; Marshall, 1890; Rosenthal & Strange, 2004). Some level of firm proximity is a necessary condition to foster increasing returns related to intermediate input effects and labor pooling. This study suggests that informality is a unique location constraint. Informal firms in this study cannot participate in urban land markets through normal (formal) profit-maximizing decisions due to price constraints or due to lack of official legitimacy. As a result, informal firms face deep restrictions on location. Conceptually, the more restrictions firms face, the less likely they will be able to locate near suppliers, markets, competitors, or generally at locations where agglomeration economies are most likely. While further research is required, results indicate, at least

conceptually, that informal firms will likely settle for suboptimal location arrangements. As an example, in Nairobi's handicraft industry existing clusters must precariously rely on fickle government intervention in land markets on their behalf. While powerful agglomeration economies are observed within these clusters, these externalities and the firms themselves are all at risk of being bulldozed in the night.

Moreover, when faced with this reality, it is possible that firms that choose to avoid disruption or the threat of disruption may also be choosing to avoid otherwise prime locations or even other similar firms even though it would be advantageous to co-locate. There is likely a tradeoff to be negotiated between "safety-in-numbers" logic and the benefits of "keeping a low profile." This is suggested by Murphy's (2006) study of Tanzanian furniture manufacturers. In Murphy's study firms that gather together are frequently harassed and those that wish to avoid harassment must locate in less advantageous locations.

Second, the connection between entrepreneurship and development can only be understood in the context of the performance of local institutions (Acs, Desai, & Hessels, 2008; North, 1990). Formal institutions and the apparatus of policy implementation are weak in the context of informal production (Roy, 2005). In fact, the weakness or absence of important formal institutions is a defining factor for firms in this study. This is expressed by the relationship of firms to urban land markets, the observed lack of trust in police and courts of law, and the context of subcontracting transactions. Contracts and contract enforcement, due process protections, access to courts of law, property protections (including intellectual property), as well as responsive government representation are not fundamental parts of everyday life for informal firms. As shown here, this limits the options firms have when dealing with problems like adverse selection or other negative externalities of agglomerated production. Further, while subcontracting is an important input effect observed here, it becomes highly problematic in the context of weak institutional coverage undermining its positive impact on cluster development. Though the observation that informal firms are excluded from formal institutions and protections is not new, it is important to note that an outcome of this reality is that informal production, due to its intuitional context, can create devastating negative externalities that undermine the benefits of agglomeration.

Third, the positive effects of labor pooling as an agglomeration economy are undermined by its structure in informality. Grimm, Kruger, and Lay (2011) estimate that informal firms face several entry barriers including capital start-up costs, monthly input costs, labor costs, and human capital or skills, and finds that capital costs tend to be negligible relative to the others. In Nairobi's handicraft sector, skill acquisition, labor costs, and monthly input barriers are all rendered negligible due to commonplace labor practices such as the reliance on probationary casual labor and widespread subcontracting. While this situation might be seen by some observers as an important benefit (Ali *et al.*, 2010; ILO, 1972) because it provides seemingly limitless basic livelihood potential, there are important complications. Skills are readily taught to anyone interested in joining the industry which removes an effective entry barrier to the industry. An outcome of this dynamic is that the industry is characterized by large numbers of small firms in which the average informal firm in the industry is an owner operator with approximately four casual workers at any time making competition intense. This does not mean that monopolistic conditions would be preferable, but respondents describe conditions approximating pure competition whereby the prevailing price is lower than the average cost to produce. This is only conceptually possible over time when new entrants continually replace those exiting the industry due to operating at a loss for too long and new entrants likewise pursue a strategy to secure a sale just to get today's bread.

In effect, there is an oversupply of labor in informality which leads to destructive forms of competition. As discussed throughout, agglomeration theory anticipates that access to skilled labor will be a benefit, but in informality anyone with skill or inclination toward a firm's industry is first a potential competitor and then a potential employee. There may be reason to rethink the assumption that this leads solely to positive outcomes because of the long-run implications of this form of competition on informal industries. The constant movement of workers toward independent livelihoods keeps firms from growing internal capacity, gaining market share, and developing competitive advantage. It removes the impetus for sustained wage premiums and holds informal manufacturing down to near subsistence level productivity. Here, it seems, concentrated informal labor leads to industry instability, widespread inefficiency, and a withering impact on agglomeration economies.

7. CONCLUSION

This article theorized that conditions found in informal spaces and production processes interfere with the mechanisms of transition in developing world production clusters articulated by Van Dijk and Sverrisson (2003). It also speculated that, if true, it may help explain why the literature on agglomeration economies in informal production is often conflicting.

The research showed that these two important lower order agglomeration effects are present and are associated with positive externalities benefiting firms. This confirms the expectation of Van Dijk and Sverrisson's typology. Here, handicraft clusters most resemble *Local Market* clusters due to the overall reliance on imitation and the apparent importance of intermediate input proximity and labor pooling. This research also showed that these agglomeration effects are also likely constrained due to the relationship between informal production and urban space in Nairobi. Further, important diseconomies of agglomeration were articulated that likely arise uniquely in contexts of informality and seem to undermine clusters, leading to the relocation of many firms.

From this research, it seems plausible that one of the reasons for stunted evolution of production clusters in the developing world may be the way lower order agglomeration economies perform in the context of informality. The mechanism of transition required to move from the type of clusters observed to *Local Network* clusters and beyond are predicated on increased specialization of labor and production compliments. However, evidence presented here suggests that the diseconomies associated with lower level agglomeration effects, as they manifest in informality, are instead confounding the types of cluster productivity that are thought to propel cluster evolution. The manifestation of intermediate inputs economies here lead to important diseconomies related to transactions, increased risk to proprietary information, and the reputation for quality of firms. The way labor pooling manifests interferes with the logic of specialized skill acquisition and the development of firms' internal capacity. Both, instead of fostering productive forms of competition, lead to devastating and pernicious environments.

Handicraft production in Nairobi is but one example of clustered informal manufacturing in the developing world. However, many of the characteristics of the industry's

organization and the informal labor market observed here are found elsewhere. As a result, the messy reality of agglomeration economies in informality may be similarly confounding development trajectories in other contexts. The evidence presented here suggests that we should re-examine production cluster performance in contexts of informality. Future studies should acknowledge that in these contexts the mechanics of lower order agglomeration economies may be a double-edged sword; that over the evolutionary history of a cluster, agglomeration economies can potentially draw firms into proximity, provide important benefits to firms and then lock them into problematic patterns that constrain further cluster evolution and the growth potential of firms.

Future research questions include whether the diseconomies and confounding influences identified here lead to similar problems with subcontracting arrangements or intermediate input patterns observed elsewhere. Also, we need to know if the same labor dynamics observed in the handicraft industry lead to similar complications in other informal contexts. For instance, as discussed above, pernicious forms of competition are widely observed in informal production contexts in the literature, but more direct study of the link between informal industry labor-training patterns, flexible labor-hiring practices, and incidents of hypercompetition are required. Further, examples of successful cluster transitions exhibiting higher order agglomeration economies should be (re)examined for how they overcame diseconomies of informality. These examples would be particularly interesting for implications on the physical and social organization of informal firms. A research agenda along these lines would help inform industrial upgrading schemes by pointing out where breakdowns in agglomeration logic are likely to occur in nascent clusters characterized by high levels of informal production and inform how these problems may be overcome.

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